Supplement 6 Selected publications reporting data on the top 10 most frequently investigated genes, proteins and metabolites

The publications are tabulated according to the direction of regulation reported ("increase of expression", "decrease of expression", "no effect of tension force on expression", and "other changes in expression").

Designations used in publications	Gene symbol	Articles reporting an increase of expression[1]		Articles reporting a decrease of expression[1]		Articles reporting no effect of tension force on expression[1]		Articles reporting other changes in expression	Total number of citations
		Gene	Protein/Metabolite	Gene	Protein/Metabolite	Gene	Protein/Metabolite		
Runx2 RUNX2 / p-RUNX2 CBFA1	*RUNX2*	Chang et al. (2017), Fujihara et al. (2010), He et al. (2019), Jiang and Hua (2016), Lee et al. (2015), Li et al. (2014), Li et al. (2015), Liu et al. (2017), Ren et al. (2015), Shen et al. (2014), Tang et al. (2012), Wada et al. (2017), Wang et al. (2019a), Wang et al. (2019b), Wei et al. (2014), Wei et al. (2015), Wu et al. (2019), Yang et al. (2018), Yu et al. (2018)	Jiang and Hua (2016), Li et al. (2014), Li et al. (2015), Shen et al. (2014), Tang et al. (2012), Wang et al. (2019b), Wei et al. (2014), Wu et al. (2019), Yu et al. (2018)	Nokhbehsaim et al. (2010), Nokhbehsaim et al. (2011)				Cho et al. (2010), Li et al. (2013), Sun et al. (2017), Sun et al. (2016), Yang et al. (2010)	**26**
ALP	*ALPP*	Chang et al. (2015), Chen et al. (2014), Fujihara et al. (2010), Jacobs et al. (2013), Jiang and Hua (2016), Konstantonis et al. (2014), Lee et al. (2015), Liu et al. (2017), Monnouchi et al. (2011), Papadopoulou et al. (2019), Shen et al. (2014), Yang et al. (2018)	Chang et al. (2017), Chen et al. (2014), Jiang and Hua (2016), Konstantonis et al. (2014), Matsuda et al. (1998), Nazet et al. (2020), Shen et al. (2014), Wei et al. (2015), Yang et al. (2006)	Yamaguchi et al. (1996)	Chiba and Mitani (2004), Yamaguchi and Shimizu (1994), Yamaguchi et al. (1996)			Nokhbehsaim et al. (2010), Qin and Hua (2016), Yamaguchi et al. (2002), Yang et al. (2010)	***24***
OCN OC Osteocalcin	*BGLAP*	Chang et al. (2015), Chen et al. (2014), Jiang and Hua (2016), Kim et al. (2007), Lee et al. (2015), Ren et al. (2015), Shen et al. (2014), Wang et al. (2019a), Wang et al. (2019b), Wei et al. (2014), Wei et al. (2015), Wu et al. (2019), Yang et al. (2016), Yang et al. (2018)	Chang et al. (2017), Chen et al. (2014), Jiang and Hua (2016), Shen et al. (2014), Wang et al. (2019b), Wei et al. (2014), Yang et al. (2006), Yang et al. (2018)					Jacobs et al. (2013), Qin and Hua (2016)	**18**
IL-1β IL1β	*IL1B*	Lee et al. (2012), Nokhbehsaim et al. (2010), Ritter et al. (2007), Shimizu et al. (1997), Wada et al. (2017)	Abiko et al. (1998), Kaku et al. (2019), Liu et al. (2017), Shimizu et al. (1994), Shimizu et al. (1995), Shimizu et al. (1997), Sun et al. (2016), Sun et al. (2017)	Nokhbehsaim et al. (2012)				Long et al. (2001), Zhao et al. (2016), Zhuang et al. (2019)	***16***

1. Increase = "Increase" or "Increase with Plateau"; Decrease = "Decrease" or "Decrease with Plateau"…
2. The citations in red font appeared more than once. - **(each citation is counted only once)**

Designations used in publications	Gene symbol	Articles reporting an increase of expression[1]		Articles reporting a decrease of expression[1]		Articles reporting no effect of tension force on expression[1]		Articles reporting other changes in expression	Total number of citations
		Gene	Protein/Metabolite	Gene	Protein/Metabolite	Gene	Protein/Metabolite		
COX2	*PTGS2*	Abiko et al. (1998), Agarwal et al. (2003), Jacobs et al. (2014), Jacobs et al. (2018), Nazet et al. (2020), Nokhbehsaim et al. (2010) (20%) , Nokhbehsaim et al. (2012) (20%) , Ohzeki et al. (1999), Shimizu et al. (1998), Suzuki et al. (2014), Wada et al. (2017), Wang et al. (2011)	Long et al. (2002), Nogueira et al. (2014)			Nokhbehsaim et al. (2012) (3%)		Nokhbehsaim et al. (2010) (3%)	**14**
OPG (Cytokine)	*TNFRSF11B*	Jacobs et al. (2013), Jacobs et al. (2015), Kanzaki et al. (2006), Kanzaki et al. (2019), Lee et al. (2015), Li et al. (2015), Liao and Hua (2013), Liu et al. (2017), Monnouchi et al. (2011), Spencer and Lallier (2009), Tsuji et al. (2004)	Jacobs et al. (2013), Jacobs et al. (2015), Kanzaki et al. (2006), Kanzaki et al. (2019), Li et al. (2015), Liao and Hua (2013), Tsuji et al. (2004)	Nogueira et al. (2014), Yang et al. (2006)	Nogueira et al. (2014), Yang et al. (2006)			Yang et al. (2010)	**14**
RANKL (Cytokine)	*TNFSF11*	Kanzaki et al. (2006), Kanzaki et al. (2019), Lee et al. (2015), Li et al. (2015), Liao and Hua (2013), Liu et al. (2017), Yang et al. (2010)	Li et al. (2015), Kaku et al. (2019)	Nogueira et al. (2014), Spencer and Lallier (2009)	Liao and Hua (2013), Nogueira et al. (2014)	Tsuji et al. (2004)		Jacobs et al. (2013), Jacobs et al. (2015), Monnouchi et al. (2011)	**14**
COL1A1 COLIα1 Collagen type-I (COL-1) Colα1 (collagen α1) COL1 COL-I	*COL1A1*	Chen et al. (2015), He et al. (2004), Lee et al. (2015), Takano et al. (2009), Wang et al. (2019b), Yang et al. (2018), Yu et al. (2018)	He et al. (2004), Qin and Hua (2016), Takano et al. (2009), Wang et al. (2019b), Yu et al. (2018)	Nemoto et al. (2010), Sun et al. (2017)	Sun et al. (2016), Sun et al. (2017)			Jacobs et al. (2013) Yamaguchi et al. (2002)	**13**
PGE_2	PGE_2		Abiko et al. (1998), Agarwal et al. (2003), Jacobs et al. (2014), Long et al. (2002), Ngan et al. (1990), Nogueira et al. (2014), Ohzeki et al. (1999), Shimizu et al. (1995), Shimizu et al. (1998), Suzuki et al. (2014), Yamaguchi et al. (1994)				Jacobs et al. (2018)		**12**
OSX SP7 Osx	*SP7*	Li et al. (2014), Li et al. (2015) Ren et al. (2015), Tang et al. (2012), Wang et al. (2019a), Wang et al. (2019b), Yang et al. (2018), Yu et al. (2018)	Chang et al. (2017), Li et al. (2014), Li et al. (2015), Tang et al. (2012), Wang et al. (2019b), Yu et al. (2018)	Li et al. (2013)				Li et al. (2013) (protein)	**10**

1. Increase = “Increase” or “Increase with Plateau”; Decrease = “Decrease” or “Decrease with Plateau”…
2. The citations in red font appeared more than once. - **(each citation is counted only once)**

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